hormone, glucagon, also made and released in the pancreas. Glucagon raises sugar levels by promoting the release of glucose from the liver. Finally, as Dr Mithal points outs, "These drugs slow the movement of food from the stomach into the small intestine. As a result, you feel full faster and longer."

Thus semaglutide not only assists the body's insulin-secreting function, it also curbs hunger, which is the reason for loss of weight. Dr Jasjeet Wasir, head of the department of endocrinology and diabetes at Medanta Hospital, Gurugram, classifies hunger as being of two types—homeostatic and hedonic (see *The Science of Hunger*). Homeostatic hunger controls balance in the body by increasing the desire to eat when energy is depleted. This usually manifests itself as weakness, grumbling in the stomach and sometimes even a headache if food is not eaten in time. Hedonic hunger is when one eats simply for the purpose of pleasure rather than any need for food. At such times, the body reaches for high-fat, high-sugar foods. Semaglutide prevents hedonic eating by promoting satiety or the feeling of fullness. And since it does not cause low blood sugar, it is useful for non-diabetics too.

The current three semaglutide preparations from Novo Nordisk differ in form and frequency. The injectable Ozempic delivers a 2 mg dose directly into the bloodstream and is recommended once a week. The FDA approved its use in Type 2 diabetes in 2017, but it has yet to be launched in India. Wegovy, too, is a weekly injectable, with a higher dose of 2.4 mg. Acquiring the FDA nod in 2021, it is the only GLP-1 agonist that has been approved specifically for weight loss treatment and, more recently, for adults with CVD. Rybelsus, on the other hand, received FDA approval for diabetes mellitus medication along with Ozempic in 2017, and comes in the form of 3 mg, 7 mg and 14 mg tablets, to be taken once daily. In the case of Eli Lilly, Mounjaro got approval in May 2022 and Zepbound last year in November.

In India, the CDSCO approved Rybelsus based on data submitted by Novo Nordisk, which sponsored a 3a trial (studies conducted after demonstrating the efficacy but before the submission of a new drug application) called Pioneer trials. Randomised controlled trials, they involved 10,000 patients worldwide, including some Indians. Adults with Type 2 diabetes not controlled by diet or exercise were randomised to receive oral semaglutide of 3 mg, 7 mg and 14 mg once daily, or a placebo. The results showed that the former exhibited better blood sugar control, weight reduction and cardiovascular safety. Later, in a 3b trial (carried out after the application of the drug but before approval), also sponsored by Novo Nordisk, diabetics were treated with 25 mg and 50 mg of oral semaglutide. Patients treated thus exhibited a significantly higher HbA1c reduction of 1.9 and 2.2 percentage points respectively, compared to the 1.5 percentage point fall in those treated with a 14 mg dosage of semaglutide. An HbA1c level between 5.7 per cent and 6.4 per cent sets you up for pre-diabetes and above 6.5 per cent puts you in the diabetic bracket. The former cohort also experienced a weight loss of 7 kg and 9.2 kg, respectively, compared to the 4.5 kg loss with lower doses.

What's the catch then, you may well ask. The problem is, we don't know yet.

NOT SUCH A SWEET STORY

Like any other drug, semaglutide, too, can have multiple side-effects. The FDA has recorded 14,000 such cases since 2018. Of these, 9,000 involved gastro-intestinal issues—nausea, vomiting, diarrhoea, upper abdominal pain and abdominal dissension. Among the severe warnings are gastroparesis (stomach paralysis), gall bladder issues, vision problems and pancreatitis. In October 2023, the list saw the inclusion of ileus, a painful condition where the intestines lose their ability to contract, thus not allowing waste to move out of the body. In extreme cases, these drugs can also lead to medullary thyroid or endocrine cancer.

"Some patients have such a strong reaction after just one tablet that they cannot bear to look at food again," says Dr Richa Chaturvedi, senior consultant, endocrinology, at Indraprastha Apollo Hospital, Delhi. "Most patients take 3 mg and somehow tolerate 7 mg." Indeed, Sushmita Bhatia (name changed), a public relations executive in Mumbai, used Rybelsus on her sister's prescription. Neither obese nor diabetic, all the 40-year-old wanted to do was to lose some weight to fit into a new dress. "I vomited for 2-3 days as soon as I had my first pill," she says. Once she stopped it, her gastric symptoms retreated in four days. Her 45-year-old sister, who was diabetic and had a BMI of 28, classified as overweight, was fine

THE NEGATIV

Like all drugs, semaglutide also has side-effects, a majority of which are gastro-intestinal



- **Minor symptoms include nausea, vomiting, diarrhoea, upper abdominal pain and abdominal dissension**
- **Severe side-effects can lead to pancreatitis, stomach paralysis, allergic reactions, gall bladder issues, vision problems**